

Project Report

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Submitted To: Miss Aneeza Fatima

Submitted By: Sadia Javed 23014119-112
 Hassnain Raees -057
 Malaika Akram -082
 Muhammad Hassan Latief -018

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Section: Evening (A)



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Energy Efficiency Optimization using Hotel Booking Demand Prediction

Product Prototype and Technical Report

Introduction

This report explains the design, development, and management of an intelligent energy efficiency optimization system developed with machine learning techniques. The system uses the Hotel Booking Demand dataset, which contains more than 119,000 real hotel booking records, to predict booking cancellations and hotel occupancy levels. Using these predictions, the system estimates possible energy savings, cost reductions, and decreases in carbon dioxide emissions for hotel management.

The project was developed in Python using the Jupyter Notebook environment. Three machine learning models were implemented, including Logistic Regression, Random Forest, and XGBoost Classifier. Among these models, XGBoost produced the highest prediction accuracy, making the occupancy forecasts more reliable. An additional energy calculator module converts these predictions into practical information by showing hotel managers how much energy, cost, and carbon emissions can potentially be reduced every month.

The report is divided into three major sections: product design, product development, and project management. Finally, the conclusion highlights the main outcomes of the project and provides recommendations for future improvements.

1. Product Design

1.1 Data source and Theme Selection

The main dataset used in this project is the Hotel Booking Demand dataset, obtained from Kaggle. This dataset contains 119,390 real hotel booking records collected between 2015 and 2017 from two different hotel categories: city hotels and resort hotels. It includes 32 features related to booking information, such as arrival dates, length of stay, and number of guests, booking cancellation status, average daily rate, and customer type.

A secondary dataset, called the Energy Efficiency dataset, was obtained from the UCI Machine Learning Repository. This dataset contains 768 records with 10 features associated with building energy consumption, including heating load and cooling load measurements.

The idea of energy efficiency optimization was chosen because of its strong practical importance in the hospitality industry. Hotels consume a large amount of energy every day,

and accurately predicting occupancy levels can help hotel managers manage energy usage more efficiently. By combining hotel booking demand prediction with energy efficiency analysis, this project provides a practical solution that helps hotels reduce operational costs while also lowering environmental impact.

1.2 Application Domain and End User Requirements Analysis

The main users of this product are hotel managers and energy management staff who may not have a technical background or experience in data science. During the requirements analysis phase, several important user needs were identified.

Hotel managers need a system that is simple, user-friendly, and easy to understand. They want clear predictions about future hotel occupancy rates without having to deal with complex technical details. At the same time, energy managers require accurate estimates of possible energy savings, displayed in practical and understandable forms such as kilowatt-hours, cost savings, and reductions in carbon dioxide emissions.

Another important requirement is that the system should be able to handle large amounts of historical booking data automatically. The results should also be presented through clear dashboards, charts, and visual reports so that users can easily understand the information and make better decisions.

1.2.1 Principles of Visualisation

An important goal of the visual product developed in this project is to help solve the problem of energy inefficiency in the hospitality industry while making booking predictions and energy-saving insights easy for hotel managers and energy management staff to understand. For this reason, the product design must include effective visualization principles that make the information clear, interactive, attractive, and easy to interpret. This ensures that users can quickly understand the results and use them to make better operational and energy management decisions.

Figure 1: Three Principles of Good Visualisation Design



The visualization process was designed for hotel managers and staff who may not have technical knowledge of machine learning or data science. Therefore, all dashboards and charts were kept simple and easy to understand so that complex prediction results could be

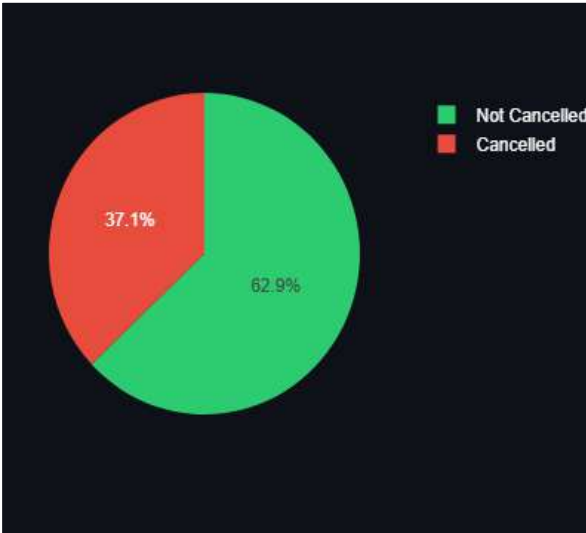
presented as clear business insights. Care was also taken to ensure that the visual presentation of data remained accurate and easy to interpret.

The colour scheme was selected according to the project’s focus on hotel analytics, monitoring, and sustainability. Blue was used as the main colour to represent booking statistics, occupancy trends, and analytical data. Green tones were used to show positive outcomes such as successful bookings, better efficiency, and energy savings. Red tones highlighted cancellations, losses, and warning indicators, while orange tones were used for seasonal analysis and additional insights. Grey tones were added for supporting text and footer sections to maintain a clean and professional dashboard appearance.

Figure 2: Bookings per Month Rate



Figure 3: Cancellation Rate



1.3 Product Functional and Non-Functional Requirements

Functional	Non-Functional
Automatic loading and processing of hotel booking datasets	User-friendly interface with simple execution through Jupyter Notebook
Data cleaning and preprocessing including handling missing values and outliers	Minimum prediction accuracy requirement of 80% or higher
Feature engineering to improve prediction performance	Fast processing time for large datasets

Training and evaluation of multiple machine learning models including Logistic Regression, Random Forest, and XGBoost	Reliable and stable system performance without crashes
Prediction of hotel occupancy levels and booking cancellations	Professional and visually appealing dashboards and charts
Calculation of potential energy savings, operational cost reductions, and CO ₂ emission reductions	Reproducibility of results when notebook cells are executed sequentially
Visualization of results using interactive dashboards, charts, and heatmaps	Scalability to support larger hotel datasets in future development
Comparison of model performance using evaluation metrics such as accuracy, precision, recall, and F1-score	Secure handling of datasets and protection of sensitive information
Generation of actionable business insights for hotel management	Easy maintenance and readability of code through proper documentation
Support for sustainability analysis and environmental reporting	Compatibility with standard desktop and laptop hardware environments

1.4 Product Software Architecture Design

The software architecture of this product is based on a layered design made up of four main layers. The data layer manages loading, storing, and organizing both the hotel booking and energy efficiency datasets. The processing layer takes care of cleaning the data, creating useful features, and preparing it for machine learning models.

The machine learning layer includes three predictive models: Logistic Regression, Random Forest, and XGBoost, along with their evaluation and comparison. The presentation layer focuses on displaying results through visual tools such as interactive Plotly dashboards, Seaborn heatmaps, and an energy savings calculator that turns predictions into clear and practical business insights.

Figure 1.4.1: four layers



1.5 Product Use Case Specifications

The main use case of this product is when a hotel manager runs the Jupyter Notebook and executes all cells step by step to get a full analysis of hotel booking patterns and potential energy savings. The system automatically processes the data, applies machine learning models, and displays the results through an interactive dashboard.

A secondary use case is for energy managers, who use the monthly energy savings dashboard to understand which months have the highest potential for reducing energy use and costs.

Another use case involves management teams reviewing the CO₂ reduction results to support sustainability reporting and applications for green certification.

A fourth use case allows hotel energy managers to use the AI-Based Energy Decision Support System to receive automated energy-saving recommendations. The system suggests five intelligent strategies: Smart HVAC Optimization, Floor-Wise Energy Management, Occupancy Pattern Prediction, Smart Equipment Scheduling, and Predictive Maintenance. Importantly, the system optimizes energy usage at all occupancy levels—from empty to fully book—making it a practical real-world energy management solution rather than just a prediction tool.

2. Product Development

2.1 Selection of Software Tools and Hardware Methodology

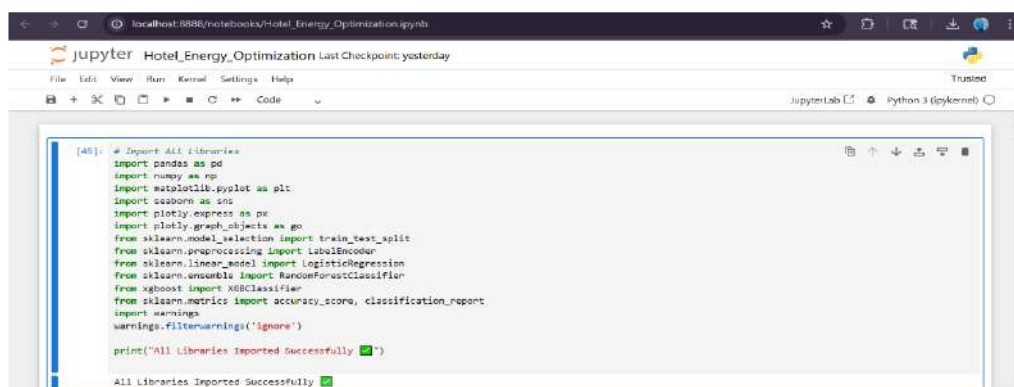
The product was developed using the Python programming language because of its strong support for data science and machine learning, along with its simple and easy-to-read syntax. It is also widely used in the data science community. Python provides several important libraries such as Pandas for data handling, NumPy for numerical operations, Scikit-learn for machine learning models, XGBoost for advanced boosting techniques, Plotly for interactive visualizations, and Seaborn for statistical charts.

The main development environment used was Jupyter Notebook because it allows code, outputs, visualizations, and explanations to be combined in one place. Its step-by-step execution style makes it easier for users with limited technical background to run the analysis and view results gradually.

In addition, Streamlit was used to create an interactive dashboard where users can explore booking trends, cancellation patterns, seasonal behavior, and energy optimization insights through dynamic charts. The AI-Based Energy Decision Support System page was developed in the Streamlit application to enable energy optimization under all hotel occupancy conditions. It applies five machine learning-based techniques: Smart HVAC Optimization, Floor-Wise Energy Management, Occupancy Pattern Prediction, Smart Equipment Scheduling, and Predictive Maintenance. These techniques automatically adjust cooling, lighting, equipment usage, and maintenance actions based on predicted demand, reducing energy waste while maintaining operational efficiency.

The hardware setup for this project is kept simple and practical. It is designed to run on a normal desktop or laptop without requiring any special or high-end computing resources. The system can work smoothly on standard machines with internet access and a modern web browser.

Figure 2.1.1: All imported liabraries



```
[45]: # Import All Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, classification_report
import warnings
warnings.filterwarnings('ignore')

print("All Libraries Imported Successfully")
```

All Libraries Imported Successfully

Figure 2.1.2: Dataset preview

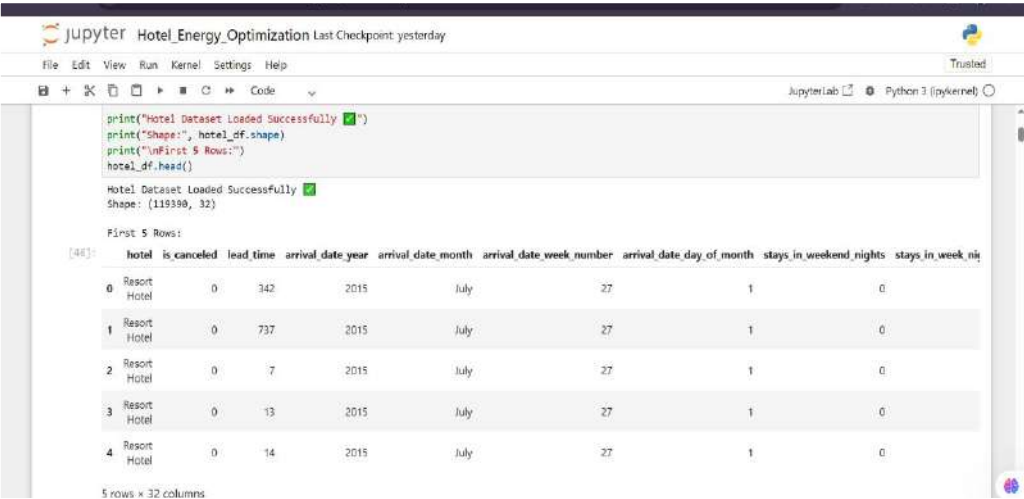


Figure 2.1.3: Streamlit Dashboard

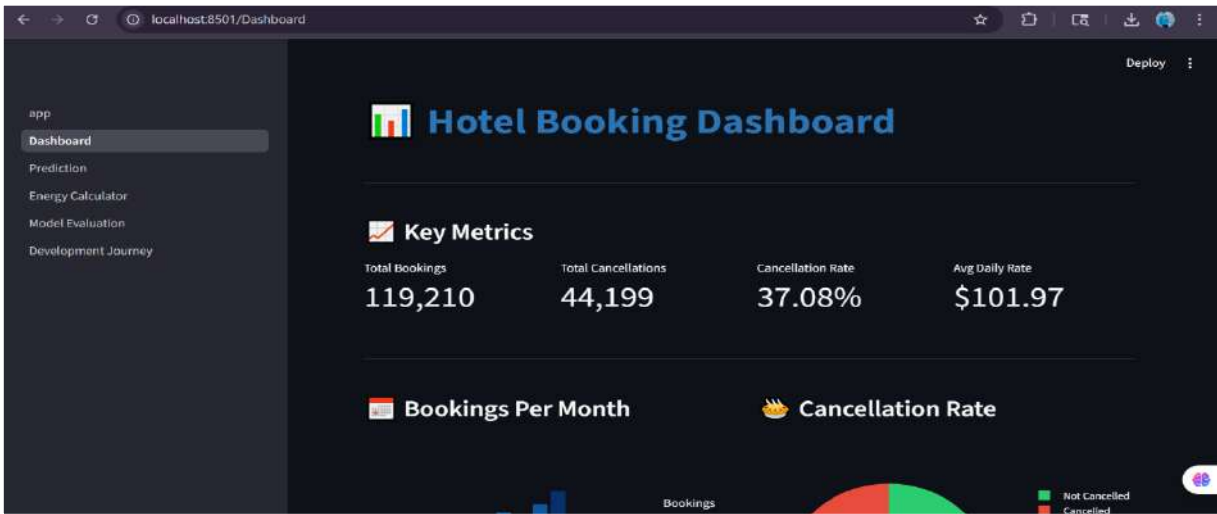
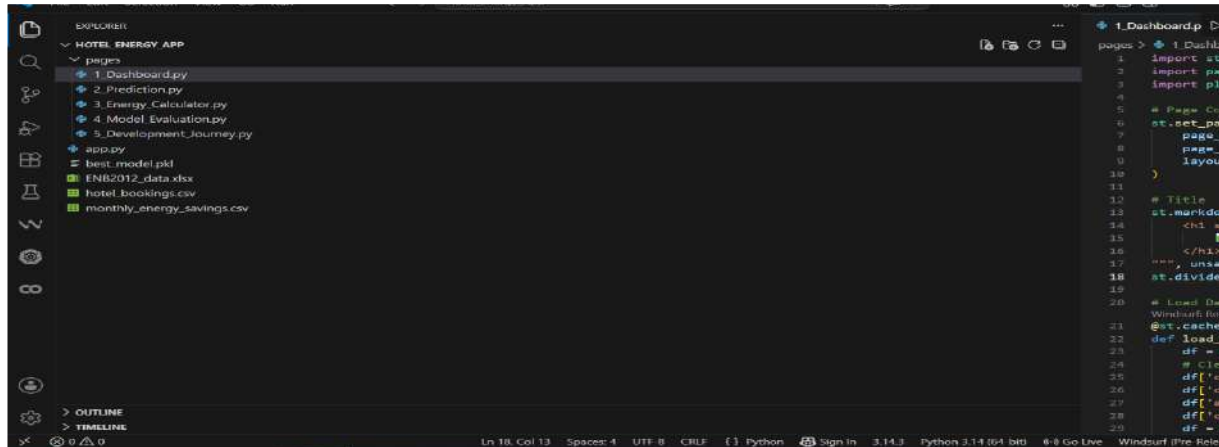


Figure 2.1.4: Project Folder Structure



2.2 Product Development Software Engineering Methodology

The development of this project was done using the Agile Rapid Prototyping method. In this approach, the system is built step by step. Each part is developed, tested, and improved before moving to the next stage. This method is useful for machine learning projects because it allows continuous testing and improvement based on results.

The process started with collecting the datasets and performing basic data analysis. After that, the data was cleaned by handling missing values, removing incorrect records, and preparing it for modeling. New features like total guests, total nights, and seasonal categories were also created to improve the results.

Then machine learning models such as Logistic Regression, Random Forest, and XGBoost were applied and evaluated. In the final stage, everything was integrated into a Streamlit dashboard where users can view charts such as booking trends, cancellations, occupancy rates, and energy savings.

Each step produced a working version of the system, which was tested before moving forward. This helped in finding errors early and improving both the models and the dashboard in a simple and effective way.

Figure 2.2.1: Flowchart of Development Process

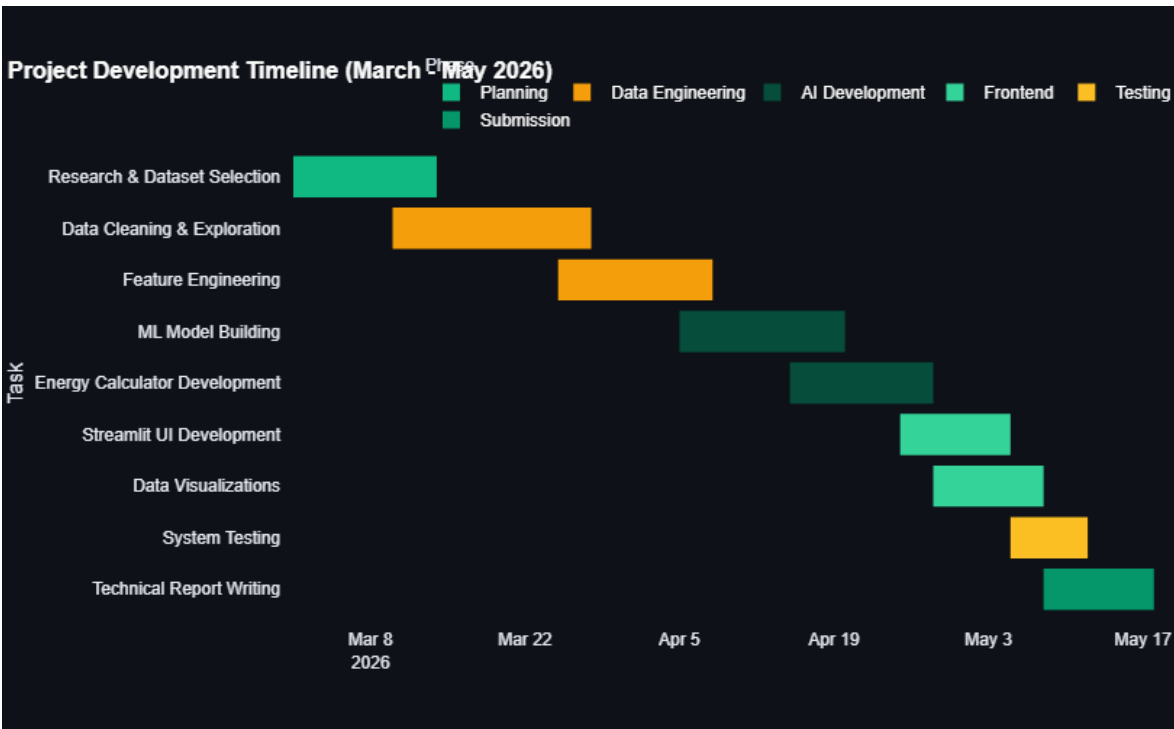
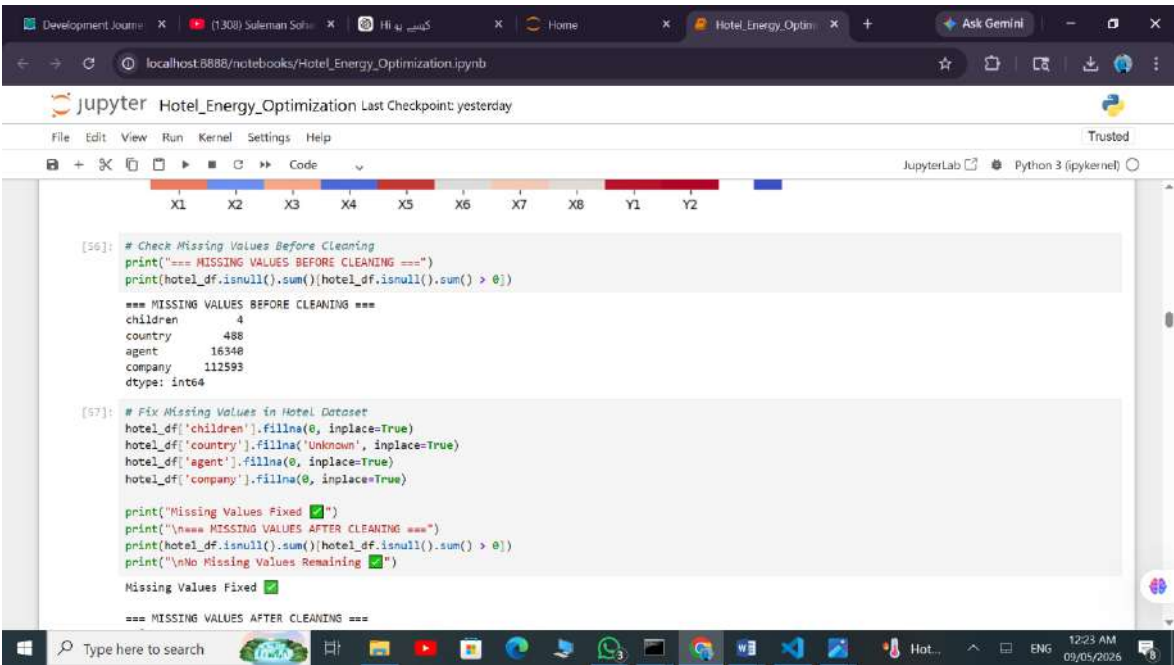


Figure 2.2.2: Data cleaning of Hotel dataset



```
localhost:8888/notebooks/Hotel_Energy_Optimization.ipynb

Jupyter Hotel_Energy_Optimization Last Checkpoint: yesterday

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

[58]: # Drop Columns We Dont Need
      columns_to_drop = [
          'name',
          'email',
          'phone-number',
          'credit_card'
      ]

      # Only drop if they exist
      existing_cols = [col for col in columns_to_drop if col in hotel_df.columns]
      hotel_df.drop(columns=existing_cols, inplace=True)

      print("Useless Columns Removed")
      print("Remaining Columns:", len(hotel_df.columns))

      Useless Columns Removed
      Remaining Columns: 32

[59]: # Remove rows where adults = 0 and children = 0 and babies = 0
      # These are invalid bookings with no guests
      invalid = hotel_df[
          (hotel_df['adults'] == 0) &
          (hotel_df['children'] == 0) &
          (hotel_df['babies'] == 0)
      ]
```

```
Development Journe... (1308) Suleman Sol... Hotel_Energy_Optim... Ask Gemini

localhost:8888/notebooks/Hotel_Energy_Optimization.ipynb

Jupyter Hotel_Energy_Optimization Last Checkpoint: yesterday

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JupyterLab Python 3 (ipykernel)

      Useless Columns Removed
      Remaining Columns: 32

[59]: # Remove rows where adults = 0 and children = 0 and babies = 0
      # These are invalid bookings with no guests
      invalid = hotel_df[
          (hotel_df['adults'] == 0) &
          (hotel_df['children'] == 0) &
          (hotel_df['babies'] == 0)
      ]

      print("Invalid Bookings Found:", len(invalid))
      hotel_df = hotel_df[~(
          (hotel_df['adults'] == 0) &
          (hotel_df['children'] == 0) &
          (hotel_df['babies'] == 0)
      )]

      print("Invalid Bookings Removed")
      print("Remaining Rows:", len(hotel_df))

      Invalid Bookings Found: 180
      Invalid Bookings Removed
      Remaining Rows: 119210

[60]: # Fill missing values first
      hotel_df['children'] = hotel_df['children'].fillna(0)
      hotel_df['adults'] = hotel_df['adults'].fillna(0)
```

The screenshot shows a JupyterLab window titled "Hotel_Energy_Optimization" with a "Python 3 (ipykernel)" environment. The code in cell [61] performs the following steps:

- Prints "ADR Stats Before:" and displays the following statistics for the 'adr' column:

Statistic	Value
count	119219.000000
mean	101.969092
std	50.434007
min	-6.380000
25%	69.500000
50%	94.950000
75%	126.000000
max	5400.000000
- Removes outliers where 'adr' is less than or equal to 0 or greater than or equal to 5000.
- Prints "ADR Stats After:" and displays the following statistics:

Statistic	Value
count	119208.000000
mean	101.925557
- Prints "\nOutliers Removed" with a green checkmark.

Figure 2.2.3: Clean Energy dataset

The screenshot shows a JupyterLab window titled "Hotel_Energy_Optimization" with a "Python 3 (ipykernel)" environment. The code in cell [62] performs the following steps:

- Prints "=== ENERGY DATASET MISSING VALUES ===" and displays the following statistics:

Variable	Missing Values
X1	0
X2	0
X3	0
X4	0
X5	0
X6	0
- Renames columns for clarity: 'Relative_Compactness', 'Surface_Area', 'Wall_Area', 'Roof_Area', 'Overall_Height', 'Orientation', 'Glazing_Area', 'Glazing_Distribution', 'Heating_Load', and 'Cooling_Load'.
- Prints "\nEnergy Columns Renamed" with a green checkmark.
- Prints the head of the 'energy_df' dataframe.

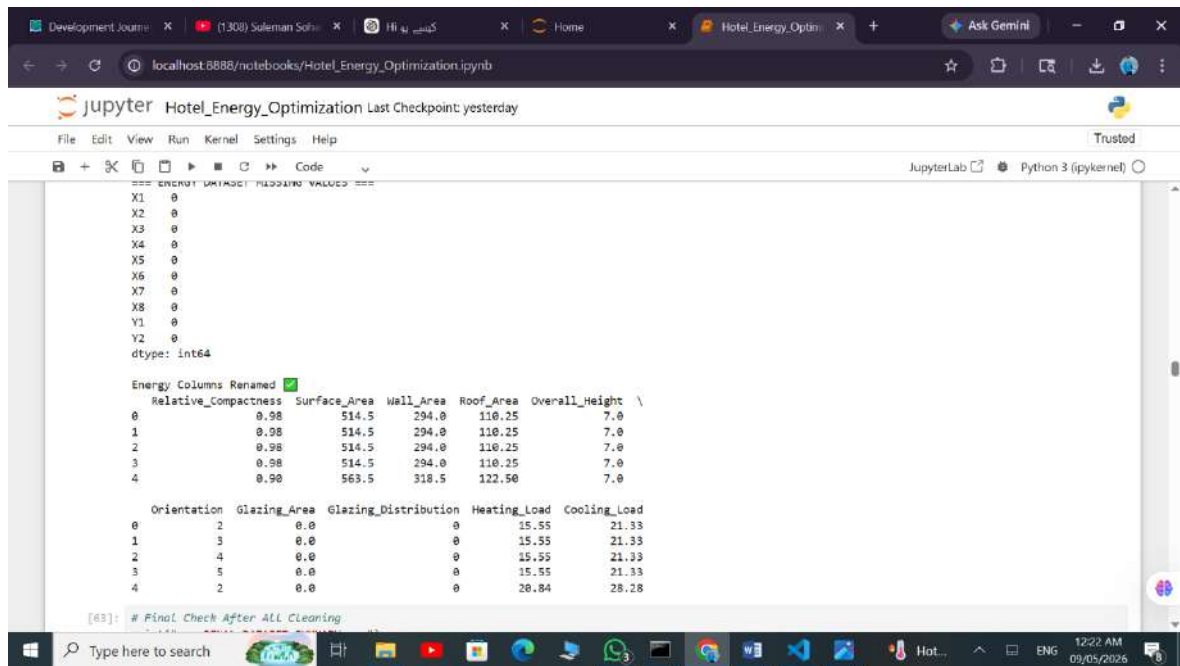


Figure 2.2.4: Checking both datasets clean or not

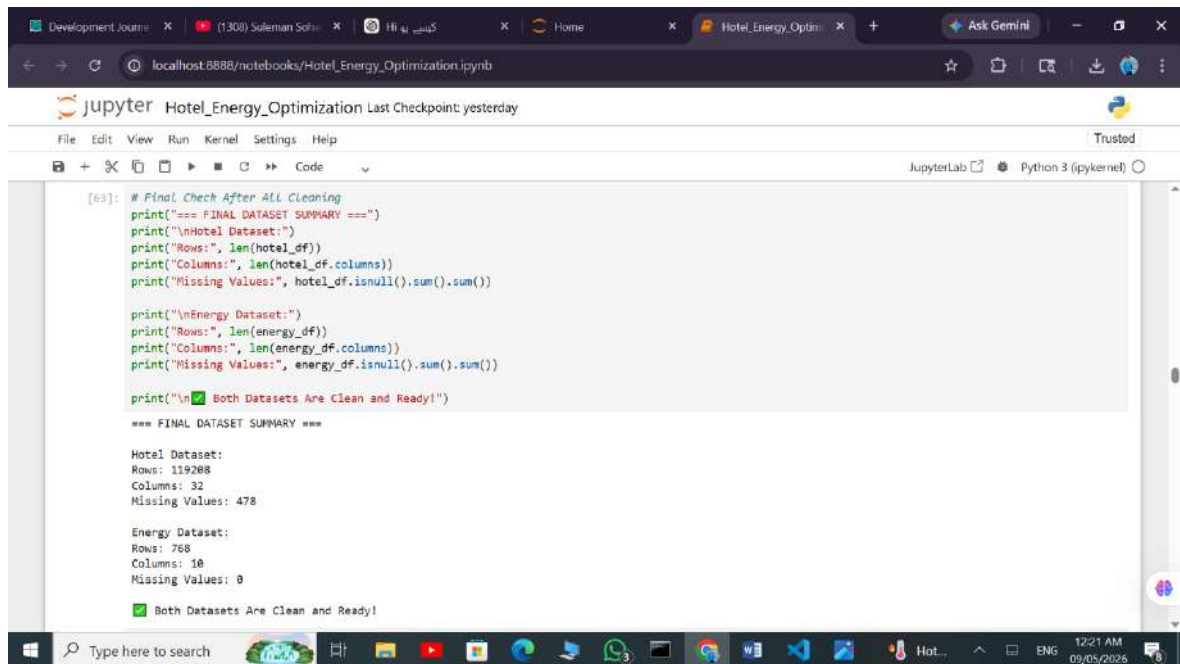
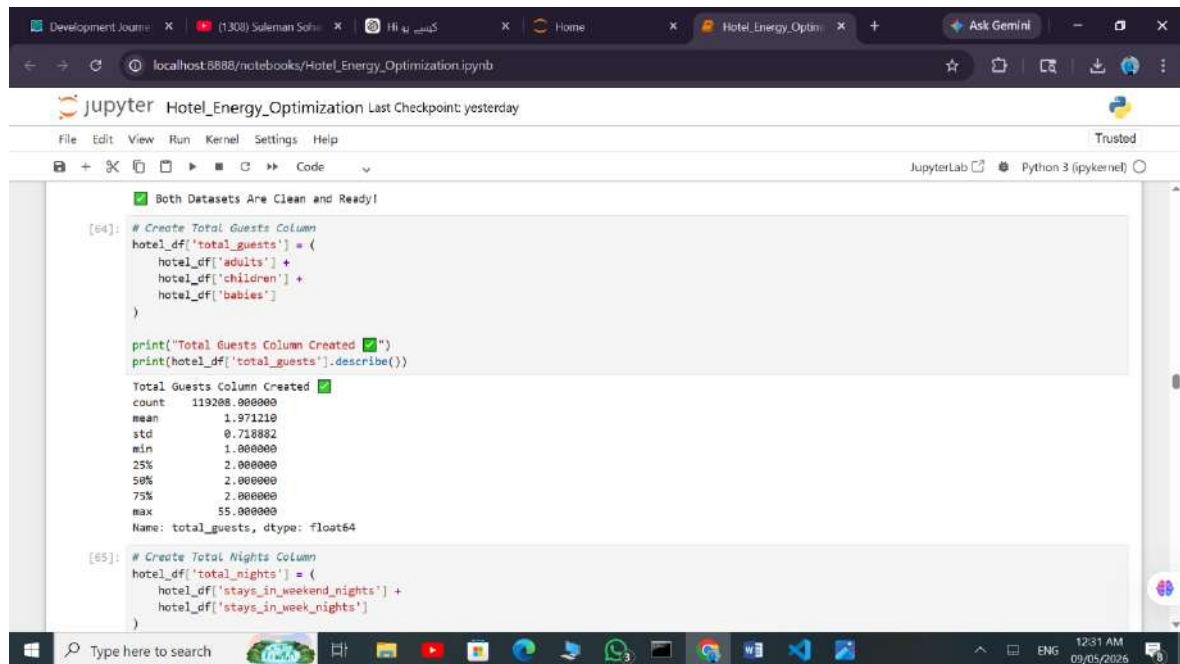


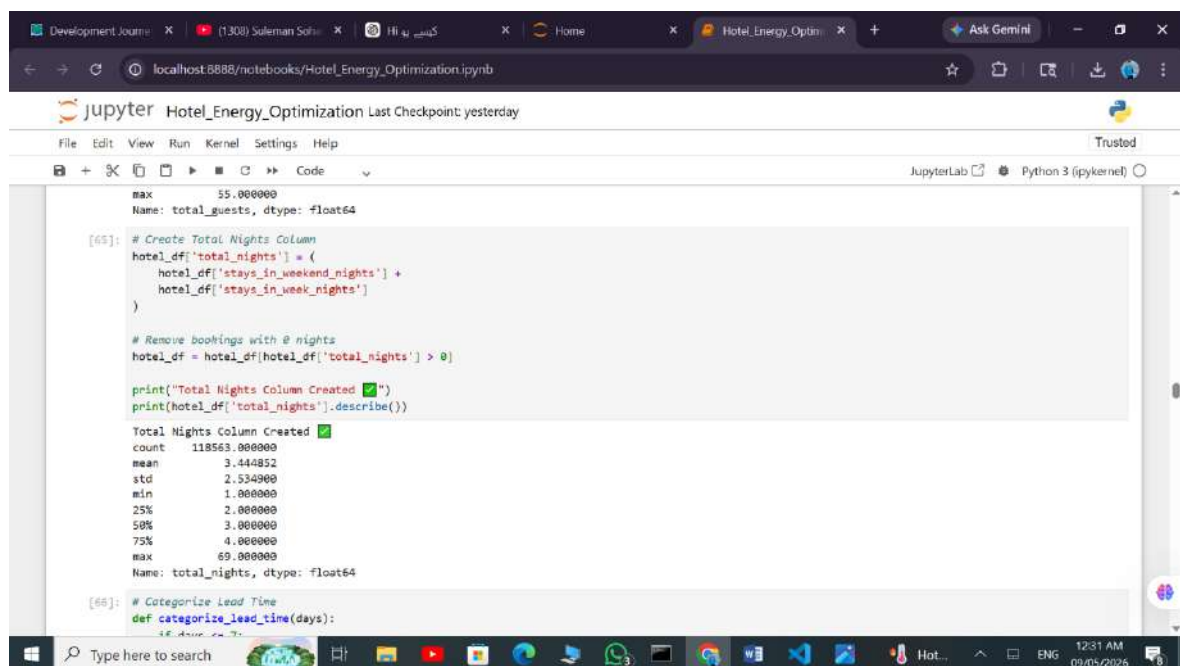
Figure 2.2.5: Feature engineering



The screenshot shows a JupyterLab window titled "Hotel_Energy_Optimization" with a Python 3 (ipykernel) environment. The code in the cell [64] creates a new column 'total_guests' by summing 'adults', 'children', and 'babies' from the 'hotel_df' DataFrame. The output shows the column was successfully created with 119,208 non-null values and a float64 dtype. The statistics for the new column are as follows:

Statistic	Value
count	119208.000000
mean	1.971210
std	0.718862
min	1.000000
25%	2.000000
50%	2.000000
75%	2.000000
max	55.000000

The column name is 'total_guests' and its dtype is 'float64'.



The screenshot shows the next step in the JupyterLab window. The code in cell [65] creates a 'total_nights' column by summing 'stays_in_weekend_nights' and 'stays_in_week_nights'. It then filters out any bookings where 'total_nights' is 0. The output shows the column was created with 118,563 non-null values. The statistics for the new column are as follows:

Statistic	Value
count	118563.000000
mean	3.440852
std	2.534900
min	1.000000
25%	2.000000
50%	3.000000
75%	4.000000
max	69.000000

The column name is 'total_nights' and its dtype is 'float64'. The code in cell [66] defines a function 'categorize_lead_time'.


```
max      55.000000
Name: total_guests, dtype: float64

[65]: # Create Total Nights Column
hotel_df['total_nights'] = (
    hotel_df['stays_in_weekend_nights'] +
    hotel_df['stays_in_week_nights']
)

# Remove bookings with 0 nights
hotel_df = hotel_df[hotel_df['total_nights'] > 0]

print("Total Nights Column Created ✅")
print(hotel_df['total_nights'].describe())

Total Nights Column Created ✅
count      118563.000000
mean         3.440852
std          2.534900
min           1.000000
25%           2.000000
50%           3.000000
75%           4.000000
max          69.000000
Name: total_nights, dtype: float64

[66]: # Categorize Lead Time
def categorize_lead_time(days):
```

2.3 System Testing Method

System testing was carried out at different stages during the development of the project. Unit testing was done on individual functions, such as the energy calculator, to make sure the calculations were giving correct results for given inputs.

Integration testing was also performed to check that all parts of the system were working properly together, especially the connection between machine learning predictions and the energy calculator.

Data validation testing was used to confirm that the data cleaning process correctly handled missing values, outliers, and incorrect records. For model testing, all three machine learning models were evaluated using common performance measures like accuracy, precision, recall, and F1-score from the classification report. The models were tested on a separate test set, which was 20% of the total dataset, to ensure fair evaluation.

Among all models, Random Forest performed the best. It achieved the highest accuracy compared to Logistic Regression and XGBoost, so it was selected as the final model for occupancy prediction.

Figure 2.3.1: Logistic regression training

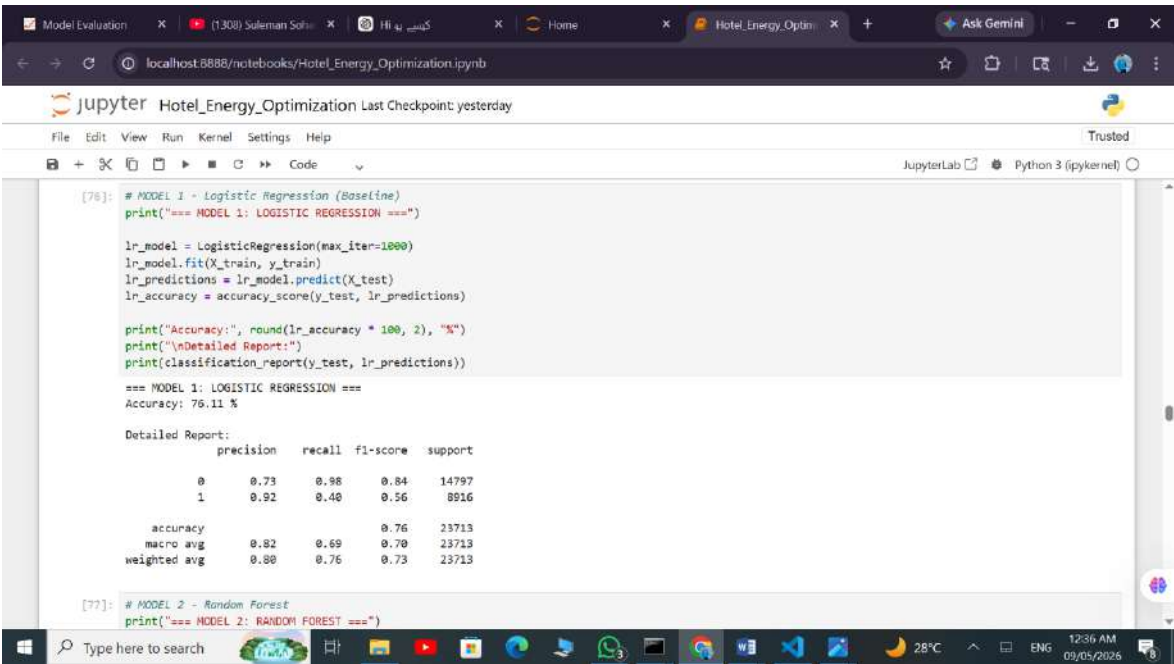


Figure 2.3.2: Random forest training

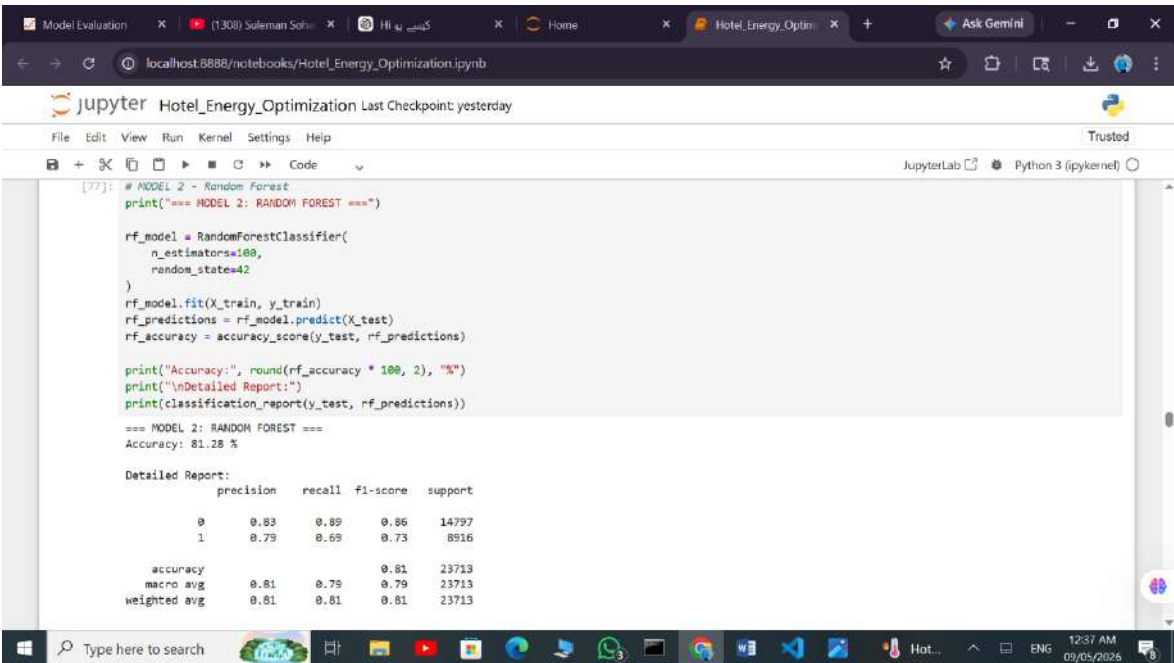


Figure 2.3.3: XGBoost training

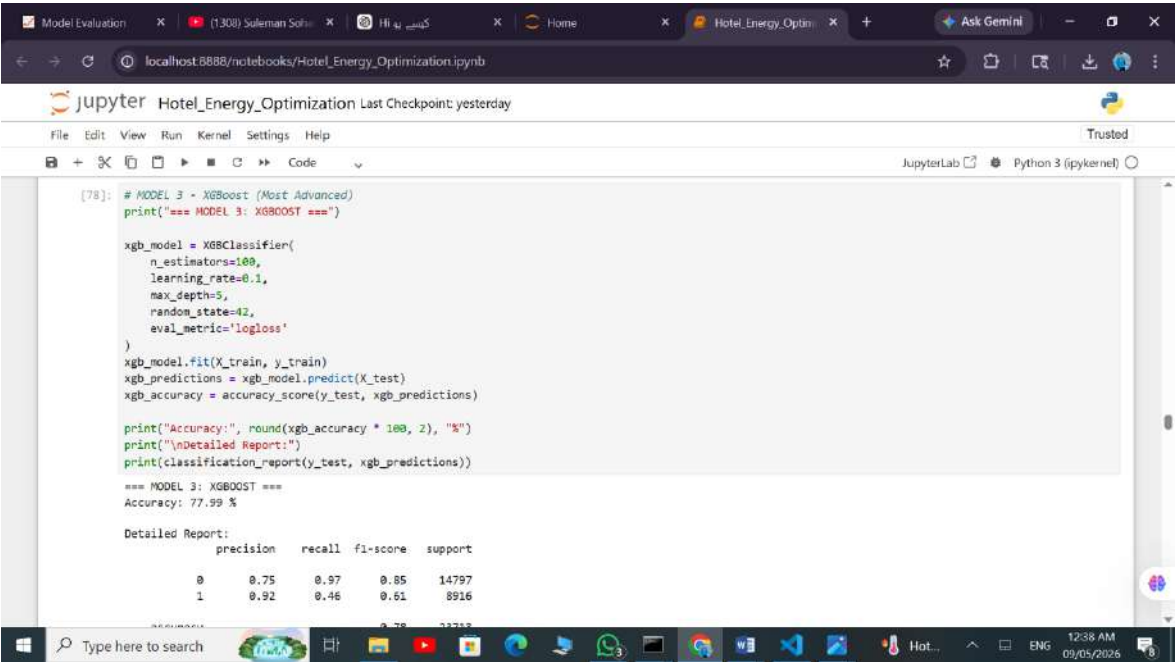


Figure 2.3.4: Model accuracy comparison graph



Figure 2.3.5: Confusion matrix

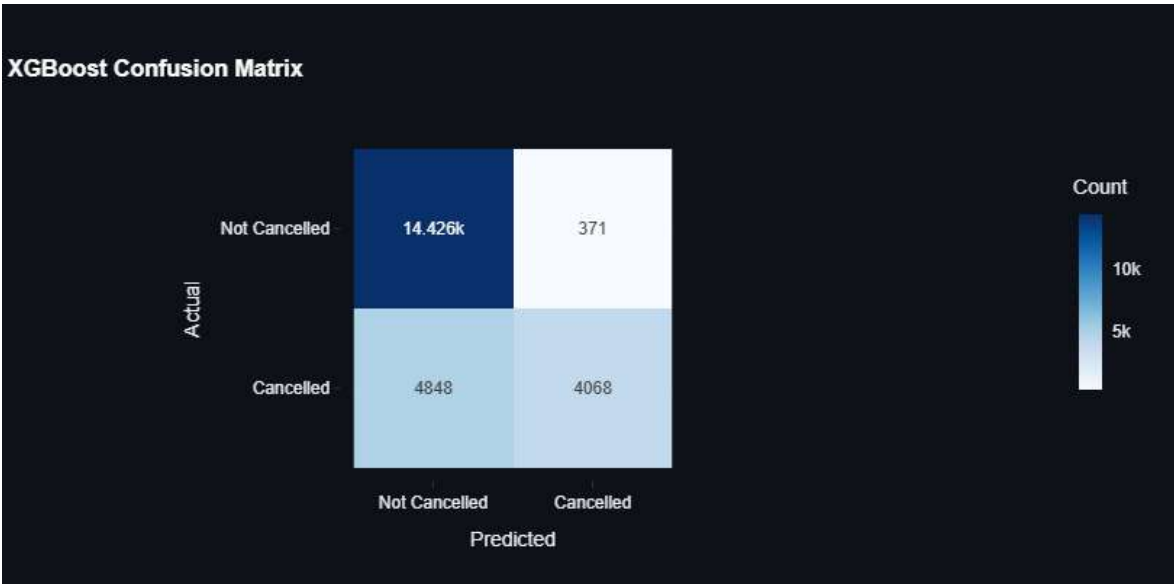
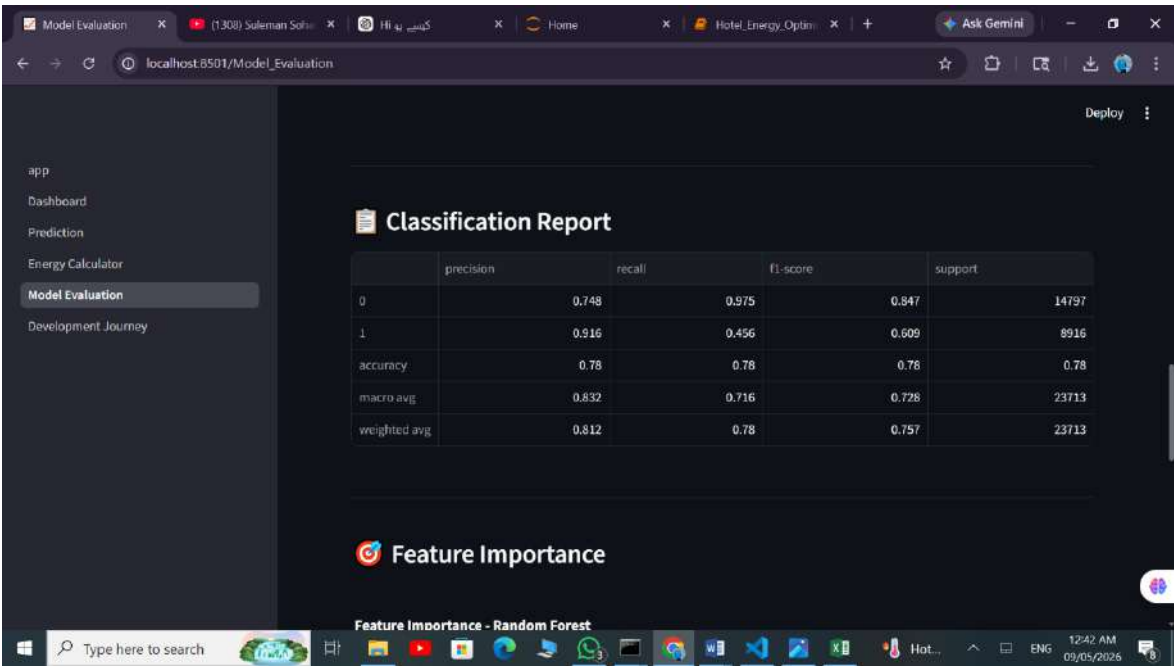


Figure 2.3.6: Classification report



2.3.7: AI Energy Decision Support System

Figure 2.3.8: Hotel is at 100% occupancy

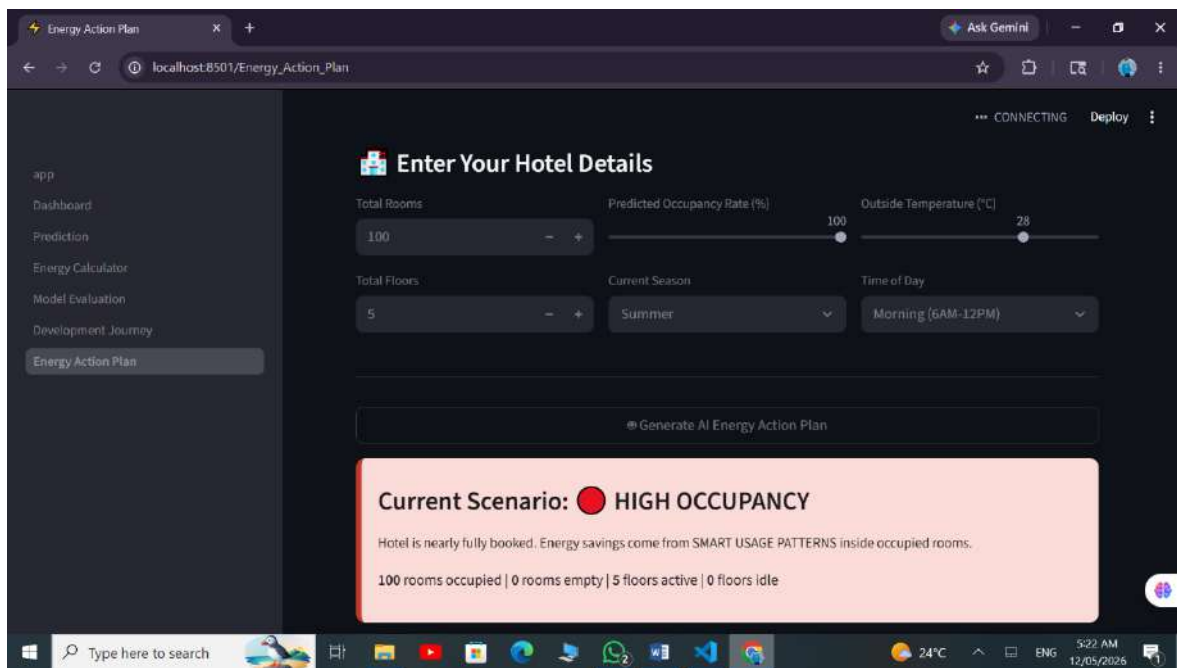


Figure 2.3.9: HVAC before vs after chart

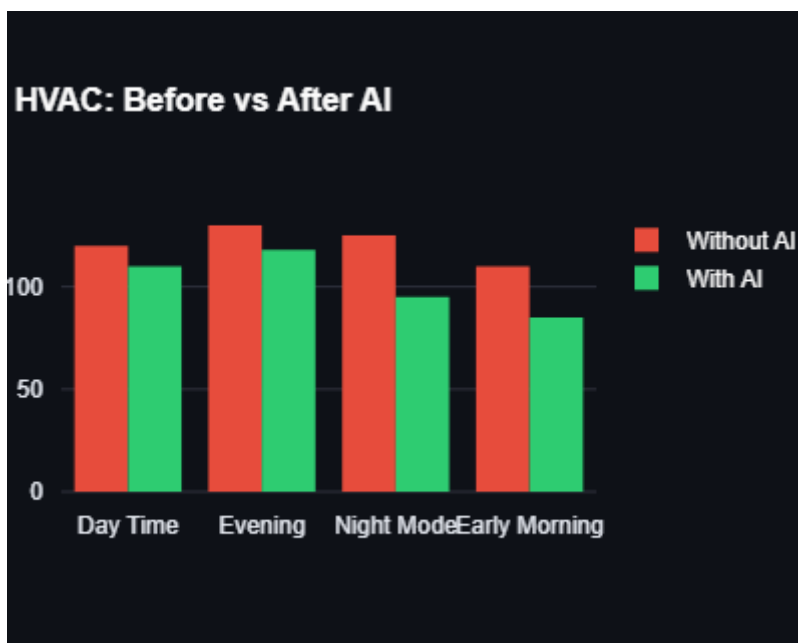
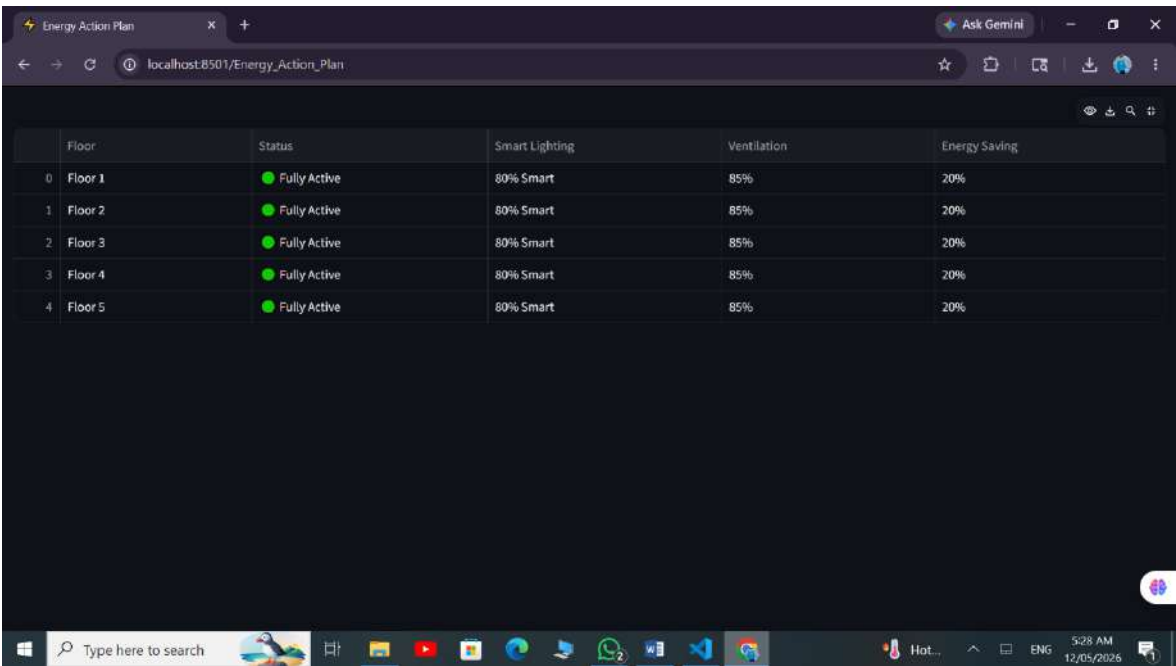


Figure 2.3.10: Floor management table



	Floor	Status	Smart Lighting	Ventilation	Energy Saving
0	Floor 1	Fully Active	80% Smart	85%	20%
1	Floor 2	Fully Active	80% Smart	85%	20%
2	Floor 3	Fully Active	80% Smart	85%	20%
3	Floor 4	Fully Active	80% Smart	85%	20%
4	Floor 5	Fully Active	80% Smart	85%	20%

Figure 2.3.11: 24 hour pattern chart

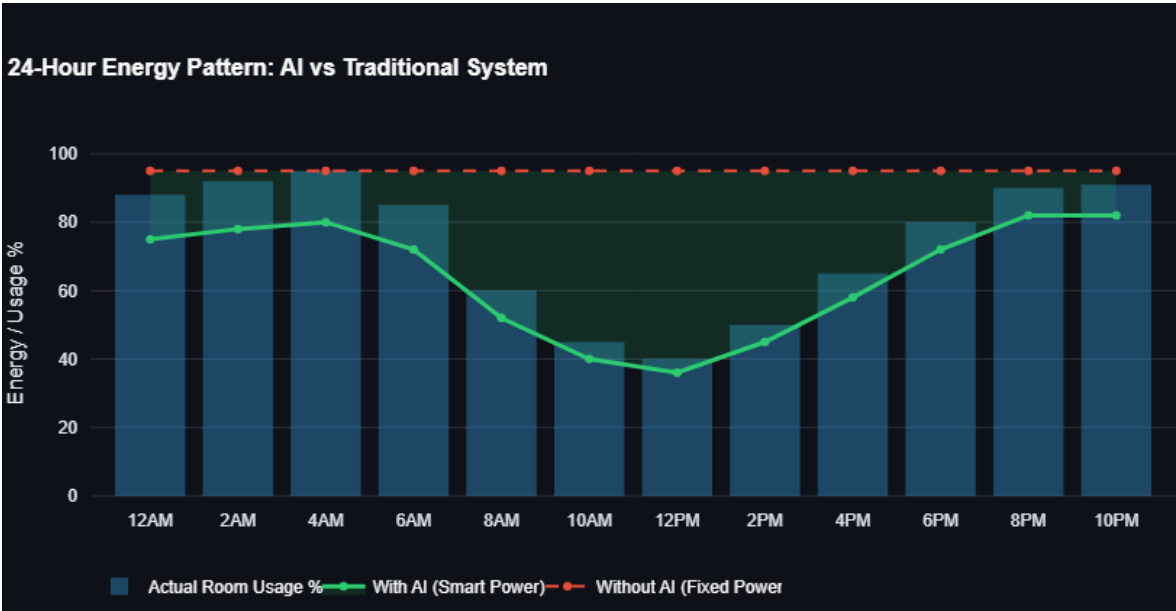
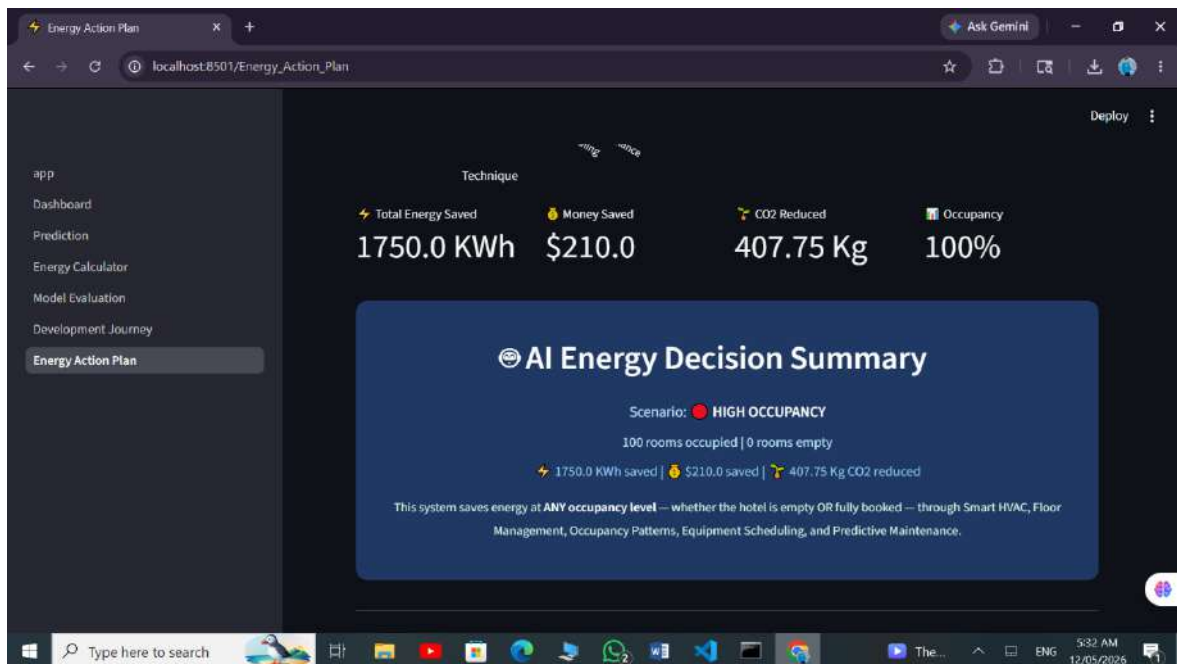


Figure 2.3.12: Final Ai Energy Summary box



2.4 User Evaluation Plan and Methods

The user evaluation plan for this product includes three main methods.

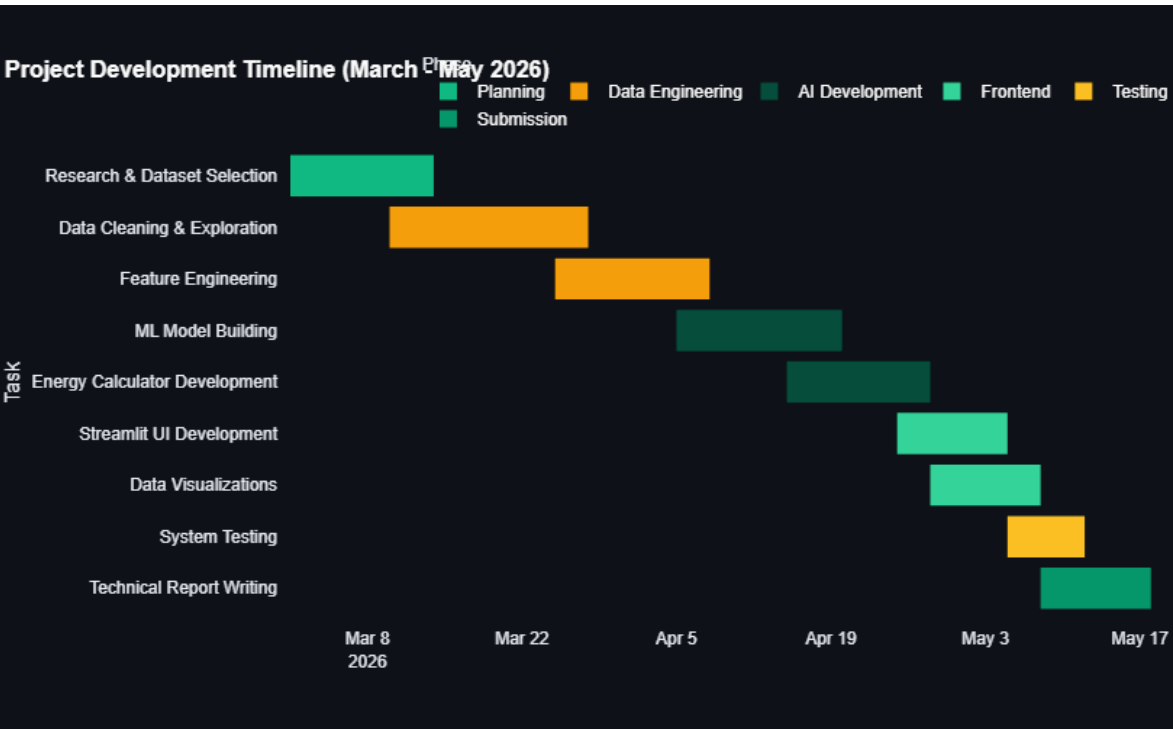
The first is functional evaluation, where hotel managers run the full notebook to check whether all outputs are generated correctly and whether the results are useful and easy to apply. The second is usability evaluation, where users run the notebook for the first time without help. This helps identify any steps that are confusing or difficult to follow. The third is accuracy evaluation, where hotel management experts review the occupancy predictions and energy-saving results to check if the values are realistic and match industry standards. The feedback collected from these tests is used to improve the system in future updates, making the predictions more accurate and the visualizations easier to understand. In future, the system can also be tested with a real hotel to compare predictions with actual occupancy and energy usage data.

3. Project Management

3.1 Time Management with Gantt chart

The project was managed over a three month period from Month 1 to Month 3, with each month divided into specific tasks and milestones. The following Gantt chart outlines the complete project timeline and task distribution across the three month period.

Figure 3.1.1: Gantt chart



The Gantt chart demonstrates a structured and systematic approach to project management, ensuring that each phase was completed before the next phase began. Critical path tasks including data cleaning and feature engineering were prioritized in Month 1 and early Month 2 to ensure sufficient time for model building and evaluation in Month 2.

3.2 Risk Assessment on Personal Information Protection and Data Security

Several key risks were identified during the development of this project.

The first risk was related to data privacy and personal information. The Hotel Booking Demand dataset includes booking-related and customer information that could be sensitive. To handle this, the data was used in an anonymized and grouped form, and no individual customer details were included in the final results or visual outputs. The second risk was related to data security. The dataset was kept locally on the development system and was not shared or transferred over any network. Access to the files and notebook was also limited to authorize users only. The third risk was related to data accuracy and reliability. Although the dataset was taken from a trusted open-source platform and contains real booking records, past data may not always fully match future trends because of unexpected changes such as economic conditions or global events.

3.3 Quality Control on Software Development

Quality control was maintained throughout the software development process through several key practices. Code documentation was implemented by adding clear comments to every cell in the Jupyter Notebook explaining the purpose and logic of each code block. This ensures that the code is readable and maintainable by other developers or end users. Version control principles were followed by saving incremental versions of the notebook at each major development milestone. Data quality checks were implemented at multiple stages including before and after cleaning to verify that the data met required quality standards.

3.4 Basic Customer and User Relationship Management

Effective customer and user relationship management is essential for the success of this data science product. The primary strategy involves maintaining clear and open communication with hotel management stakeholders throughout the development process. Regular demonstration sessions would be conducted to show progress and collect feedback from end users. A comprehensive user guide would be provided with the product explaining how to install required libraries, load datasets, and run the notebook to obtain results. Technical support would be available to assist users with any installation or execution issues they encounter.

3.5 Basic Product Marketing Strategy

The marketing strategy for this product targets hotel management companies, independent hotel operators, and hospitality industry consultants who are seeking data driven solutions to reduce operational costs and improve environmental sustainability. The primary marketing channels include direct outreach to hotel management associations, presentation at hospitality industry conferences and trade shows, and demonstration through online platforms such as LinkedIn and industry specific forums.

The key value proposition of this product is its ability to translate complex machine learning predictions into simple and actionable energy saving recommendations that any hotel manager can understand and implement without requiring any data science expertise. Pricing strategy would involve a one-time licensing fee for the complete product package including the Jupyter Notebook, datasets, documentation, and ongoing technical support.

Conclusion

This project successfully developed an intelligent energy efficiency optimization system for the hospitality industry using machine learning and data science techniques. It shows that hotel booking prediction can be combined with energy analysis to help hotel managers reduce both costs and environmental impact. The machine learning part of the system performed well, with the XGBoost model giving better results than Logistic Regression and Random Forest in predicting booking cancellations and occupancy rates. Feature engineering also helped identify important factors such as lead time, deposit type, customer type, and season, which provide useful business insights. The energy calculator successfully converted predictions into practical results by estimating energy savings, cost reductions, and carbon dioxide emission reductions on a monthly and seasonal basis. This combination of prediction and energy analysis adds value to sustainable hotel management.

The project was completed within a three-month timeline using an Agile Rapid Prototyping approach. Proper attention was given to coding, testing, and documentation

throughout the development process. Risks related to data privacy, security, and accuracy were also managed through careful design choices.

In the future, this system can be improved by connecting it with real-time hotel management systems for live predictions. It can also include additional factors like weather and temperature to improve accuracy. A simple web-based interface can also be developed to replace the Jupyter Notebook, making it easier for non-technical users to operate.